

Supplementary Information

Sensitivity-aware inspection prioritisation for archaeological landscapes: a reproducible Taxila observatory

1 Scope and evidence generations

The main manuscript is the anonymous source in `main.tex`. This supplement distinguishes historical frozen records from the corrected annual bootstrap and the new seven-family spatial extension. Counts and component scores can be reproduced from frozen inputs; historical stochastic records remain labelled by their uncertainty design. No outer proxy agreement validates monument condition. The previously examined outer stripe makes the new comparison retrospective.

2 Component inventory and analytical support

Saraikala is preserved without invented geometry. Coordinates are inventory points, not property polygons. The same 17 mapped components are compared at all three radii.

Table S1: Complete Taxila component inventory and analytical geometry status.

ID	Component	Coordinate status	Longitude	Latitude	Included
139-001	Khanpur Cave	official_web_point	72.909294	33.799200	Yes
139-002	Saraikala, prehistoric mound	official_id_geometry_missing			No
139-003	Bhir Mound	official_web_point	72.819331	33.743944	Yes
139-004	Sirkap (fortified city)	official_web_point	72.829878	33.753403	Yes
139-005	Sirsukh (fortified ruined city)	official_web_point	72.849739	33.772256	Yes
139-006	Dharmarajika stupa and monastery	official_web_point	72.842667	33.744669	Yes
139-007	Khader Mohra (Akhuri)	official_web_point	72.860722	33.760806	Yes
139-008	Kalawan group of buildings	official_web_point	72.839753	33.746292	Yes
139-009	Giri complex of monuments	official_web_point	72.883325	33.729089	Yes
139-010	Kunala stupa and monastery	official_web_point	72.831444	33.750553	Yes
139-011	Jandial complex	official_web_point	72.828914	33.764456	Yes
139-012	Lalchak and Badalpur Buddhist stuppa	official_web_point	72.868019	33.781803	Yes
139-013	Mohra Moradu stupa and monastery	official_web_point	72.861053	33.761097	Yes
139-014	Pippala stupa and monastery	official_web_point	72.866686	33.765922	Yes
139-015	Jaulian stupa and monastery	official_web_point	72.875311	33.765042	Yes
139-016	Lalchak mounds	official_web_point	72.849469	33.774450	Yes
139-017	Buddhist remains around Bhallar stupa	official_web_point	72.825358	33.813497	Yes
139-018	Giri Mosque and tombs	official_web_point	72.880522	33.727769	Yes

Note: Coordinates are WGS 84 point locations transcribed from the official component map. Saraikala remains in the inventory but is excluded from coordinate-dependent analysis. Analytical circles are not official property polygons or legal buffers.

3 Corrected annual climate analysis

The notebook now resamples detrended residual blocks. A trend-restored ensemble supplies confidence intervals; a constant-baseline ensemble supplies null slopes. The 12 annual metrics form the multiplicity family. This differs from the historical annual/seasonal family, so adjusted values must not be interchanged. Heavy precipitation counts days exceeding the 1991–2020 wet-day 95th percentile (wet days have at least 1 mm rain); hot days exceed the baseline maximum-temperature 95th percentile. Heatwave duration is the longest run above that temperature threshold, without a separate minimum-duration criterion. Dry-spell duration counts consecutive days below 1 mm. Rx1day and Rx5day are the largest one-day and rolling five-day precipitation totals. The observed precipitation point slope is unchanged; the earlier notebook’s raw-series block intervals and probabilities are superseded. Validation checks newly computed probability bounds and finite ordered intervals, in addition to historical-reference checks.

Table S2: Corrected annual residual-block trends; 2,000 draws and five-year blocks.

Metric	Slope/year	CI low	CI high	p	Annual q
precipitation_sum_mm	-10.3037	-17.9150	-2.6679	0.0090	0.0270
rx1day_mm	-0.6233	-1.5426	0.4034	0.1549	0.2324
rx5day_mm	-1.7354	-3.3891	-0.2019	0.0395	0.0948
heavy_precipitation_days	-0.1429	-0.2266	-0.0582	0.0015	0.0180
consecutive_dry_days	0.1667	-0.1417	0.4322	0.2029	0.2597
temperature_mean_c	0.0418	0.0099	0.0731	0.0065	0.0260
annual_or_period_max_tmax_c	0.0333	-0.0079	0.0778	0.1204	0.2065

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Table S2: Corrected annual residual-block trends; 2,000 draws and five-year blocks.

Metric	Slope/year	CI low	CI high	p	Annual q
hot_days	0.1818	-0.1728	0.5227	0.3063	0.3342
heatwave_max_days	0.0323	-0.0449	0.1144	0.3493	0.3493
relative_humidity_mean_percent	-0.1311	-0.2852	0.0255	0.1124	0.2065
soil_moisture_mean_m3m3	-0.0013	-0.0022	-0.0003	0.0065	0.0260
et0_sum_mm	2.2850	-1.2405	5.5941	0.2164	0.2597

Precipitation and evapotranspiration slopes are in mm year⁻¹, temperature slopes in °C year⁻¹, humidity in percentage points year⁻¹, soil moisture in m³ m⁻³ year⁻¹ and event counts in days year⁻¹. Effective-sample-size diagnostics use detrended lag-one correlation and are bounded by the observed record length; they do not replace the bootstrap.

4 Ranks and uncertainty scope

The following tables retain the archived component-score and uncertainty generations. The weight experiment perturbs the integrated domain architecture; the spatial experiment resamples 150 m spectral blocks with terrain fixed. They must not be described as the same probability distribution.

Table S3: Frozen component scores and scale-dependent ranks.

Component	Rank 250 m	Rank 500 m	Rank 1,000 m	Score 500 m
139-001	15.0000	14.0000	11.0000	0.4309
139-003	14.0000	17.0000	17.0000	0.3397
139-004	10.0000	12.0000	15.0000	0.4799
139-005	8.0000	9.0000	10.0000	0.5015
139-006	3.0000	5.0000	6.0000	0.5363
139-007	7.0000	6.0000	3.0000	0.5353
139-008	12.0000	7.0000	9.0000	0.5191
139-009	4.0000	1.0000	2.0000	0.5912
139-010	5.0000	10.5000	13.0000	0.5010
139-011	16.0000	16.0000	16.0000	0.3485
139-012	17.0000	15.0000	12.0000	0.3922
139-013	11.0000	4.0000	5.0000	0.5392
139-014	6.0000	10.5000	8.0000	0.5010
139-015	2.0000	3.0000	1.0000	0.5676
139-016	13.0000	13.0000	14.0000	0.4461
139-017	9.0000	8.0000	7.0000	0.5172
139-018	1.0000	2.0000	4.0000	0.5725

Table S4: Archived spatial-block rank uncertainty; terrain held fixed.

Component	Median rank	Low	High	Top-3 frequency
139-017	1.0000	1.0000	5.0000	0.9110
139-009	2.0000	1.0000	3.0000	0.9855
139-018	3.0000	1.9750	5.0000	0.6150
139-015	4.0000	2.0000	8.5000	0.2940
139-006	6.0000	3.0000	11.0000	0.0385
139-013	8.0000	3.0000	15.0000	0.0510
139-014	8.0000	3.5000	12.0000	0.0245
139-010	8.0000	4.5000	12.0000	0.0055
139-007	9.0000	3.0000	15.0000	0.0465
139-008	9.0000	4.5000	14.0000	0.0040
139-005	9.0000	5.5000	13.0000	0.0005
139-004	10.0000	6.0000	13.0000	0.0010

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Table S4: Archived spatial-block rank uncertainty; terrain held fixed.

Component	Median rank	Low	High	Top-3 frequency
139-016	14.0000	11.0000	16.0000	0.0000
139-011	14.0000	12.0000	16.0000	0.0000
139-001	14.0000	8.5000	16.0000	0.0005
139-003	16.0000	13.0000	16.0000	0.0000
139-012	17.0000	16.0000	17.0000	0.0000

Table S5: Archived 50,000-draw integrated-domain weight sensitivity.

Component	Median rank	Low	High	Top-3 frequency
139-009	1.0000	1.0000	9.0000	0.7740
139-018	3.0000	2.0000	11.0000	0.6794
139-015	4.0000	2.0000	7.0000	0.4404
139-013	5.0000	4.0000	11.0000	0.0000
139-006	5.0000	2.0000	8.0000	0.2189
139-007	6.0000	5.0000	9.0000	0.0000
139-008	7.0000	2.0000	9.0000	0.2542
139-017	8.0000	1.0000	15.0000	0.3054
139-005	9.0000	3.0000	11.0000	0.1017
139-014	10.0000	7.0000	13.0000	0.0000
139-010	10.0000	6.0000	10.0000	0.0000
139-004	12.0000	1.0000	15.0000	0.2260
139-016	13.0000	4.0000	16.0000	0.0000
139-001	14.0000	12.0000	15.0000	0.0000
139-012	15.0000	12.0000	17.0000	0.0000
139-011	16.0000	13.0000	17.0000	0.0000
139-003	16.0000	14.0000	17.0000	0.0000

5 Spatial benchmark protocol and complete diagnostics

The input SHA-256 is 3188160a144c5020e7d21bb50a195e07a686d941134ade35a57232fe41ddc1b1. The outer development/buffer/test counts are 12,094/4,467/4,188. The minimum observed inter-sample distance is 2,040 m; the earlier nominal grid-boundary separation was 2,010 m. Both refer to the same geographic split but different distance definitions.

Inner validation covers block rows 0–2, 3–6 and 7–9, with adjacent block rows excluded from training. One fold has no water-class training observations. The six-class scoring denominator remains fixed. All model families receive three configurations; there is no test-driven hyperparameter search. Random forest and extra trees use 300 trees; histogram boosting 250 iterations; XGBoost and CatBoost 300 iterations; MLP a (64,32) hidden layout and 400 iterations. Complete parameters and per-fold scores are in the executable notebook and JSON search record. Early stopping is disabled. XGBoost and CatBoost are strong additional comparators, not evidence of a universal SOTA result.

Table S6: Paired 2,000-block-bootstrap macro-F1 differences: primary random forest minus comparator. Intervals are unadjusted exploratory comparisons.

Comparator	F1 difference	CI low	CI high
Logistic	0.0145	0.0028	0.0283
ExtraTrees	-0.0049	-0.0099	0.0014
HistGradientBoosting	-0.0248	-0.0663	-0.0117
MLP	-0.0281	-0.1063	0.0504
XGBoost	-0.0054	-0.0587	0.0094
CatBoost	0.0290	-0.0051	0.0990

Table S7: Development-only temperature calibration evaluated on the outer stripe.

Variant	Temperature	Log loss	Brier	ECE
raw	1.0000	0.4977	0.2689	0.0143
OOF-temperature	1.6583	0.5735	0.2999	0.1357

Table S8: Fixed-capacity diagnostic controls; not alternative model-selection candidates.

Control	Macro-F1	Agreement	Log loss	ECE
bands-only	0.8245	0.7954	0.5619	0.0513
indices-only	0.6706	0.6566	0.8156	0.0337
location-only	0.1404	0.2290	1.9122	0.1467
shuffled-development-labels	0.0699	0.1538	1.7117	0.0627
class-prior	0.0495	0.1743	1.7062	0.0410

Table S9: Three-seed sensitivity of the development-selected random forest.

Seed	Macro-F1	Agreement	Log loss	ECE
311.0000	0.8214	0.8128	0.4977	0.0143
733.0000	0.8214	0.8128	0.5002	0.0142
2026.0000	0.8205	0.8116	0.4987	0.0133

6 Historical reproduction discrepancies

The original frozen targets are macro-F1 0.830795, agreement 0.833095 and ECE 0.014280. The historical exact-package Windows rerun reported 0.830072, 0.828080 and 0.015117. The original bootstrap had also changed the macro-F1 denominator when a resample lacked a class; the fixed-six-class correction produced the previously audited interval 0.701–0.851. Historical expected values were not overwritten after observing a rerun. Full current execution status and environment are preserved in the accompanying receipt. Engineering validation that reuses frozen model tables is not a full model refit.

7 Additional evidence and reproduction boundary

The frozen archive contains retained scene identifiers, acquisition-window weather, seasonal climate comparisons, sensor composition, common-support counts, threshold sensitivity, harmonisation sensitivity, complete factor/domain ablations, all component ranks and source licences. The notebook additionally generates shared-block, positional and joint-uncertainty analyses. The standalone legacy integrated script references missing intermediate reflectance inputs and is not presented as a self-contained source-to-result pipeline. Original satellite data should be recovered through the acquisition manifests where needed.

The selected supplementary figures below preserve archived non-primary evidence and retain their original scientific scope. Files and caption sources are indexed in the review package.

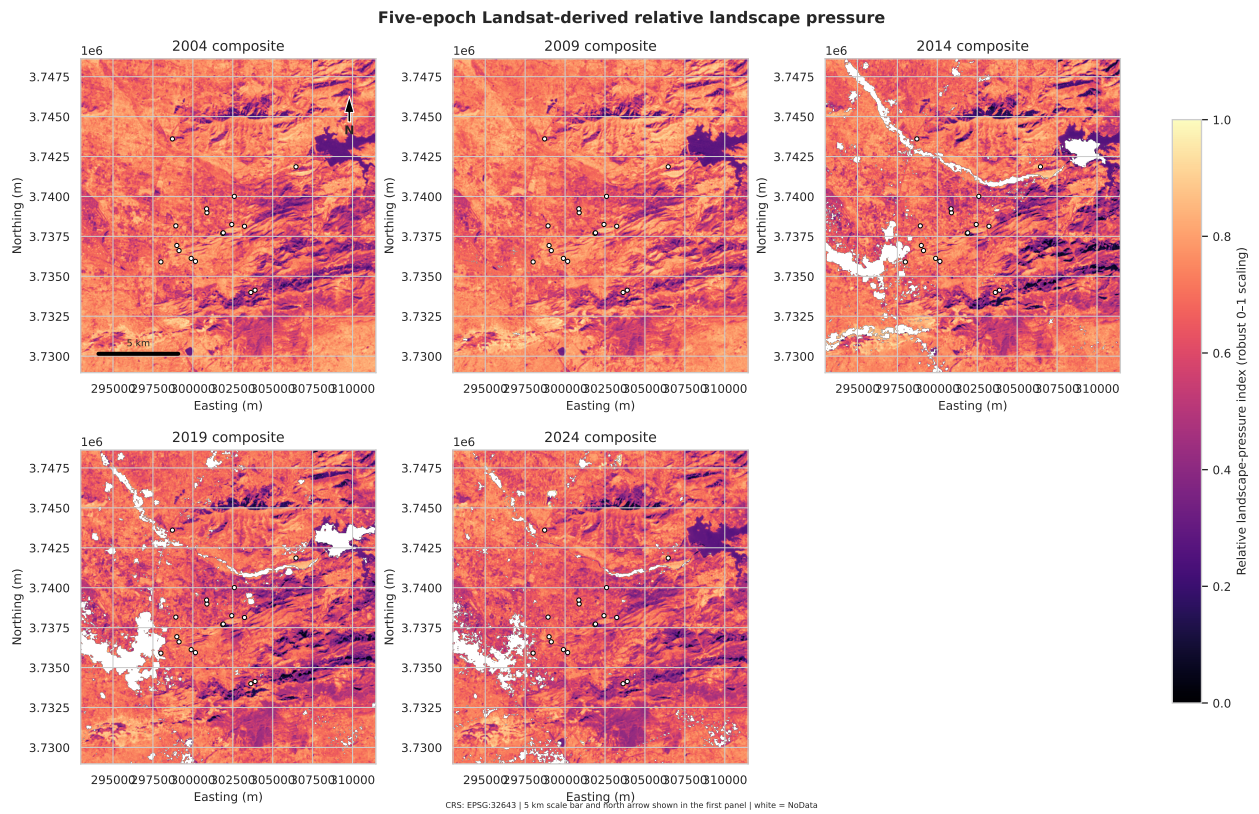


Figure S1: Five-epoch landscape-pressure context. Colour denotes relative pressure, not accumulated deterioration.

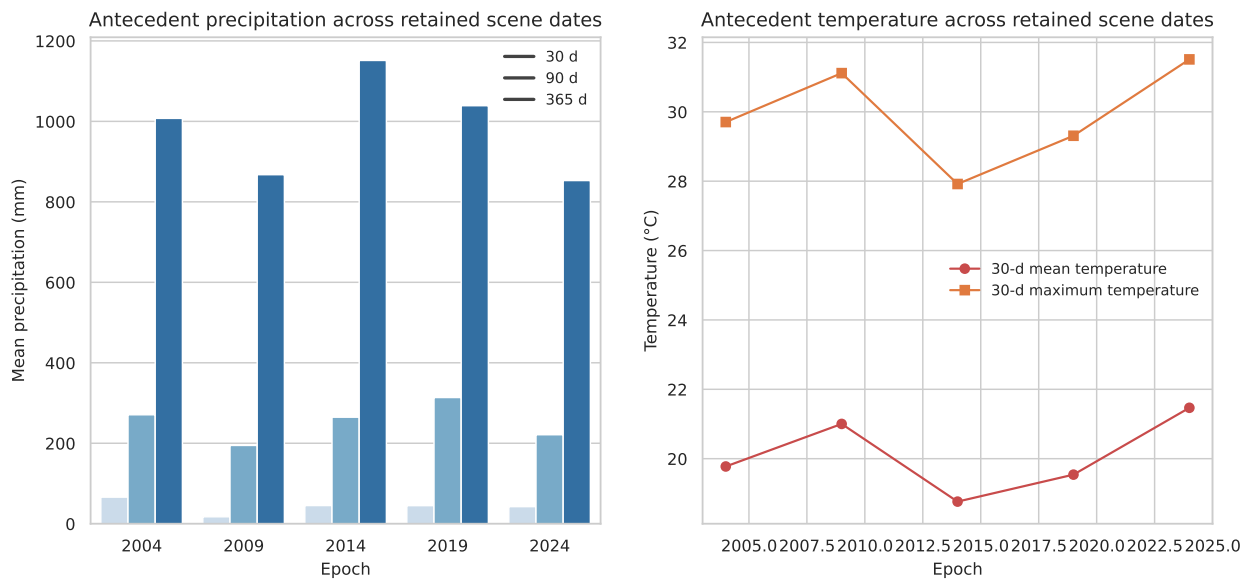


Figure S2: Weather matched to the retained Landsat acquisition windows; contextual rather than causal.

Terrain and hydrological susceptibility context

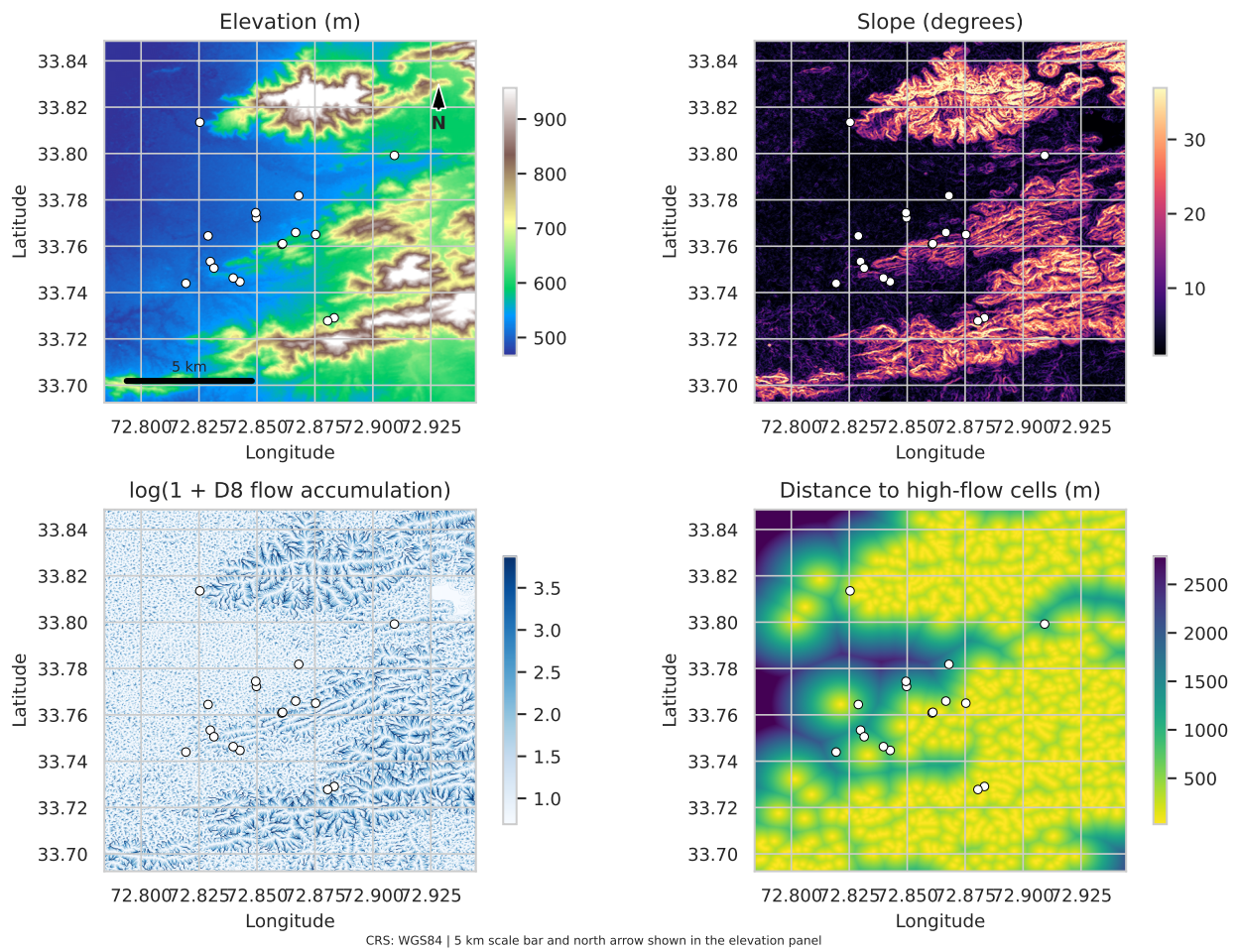


Figure S3: Elevation-derived terrain and drainage proxies. These do not constitute a hydraulic model.

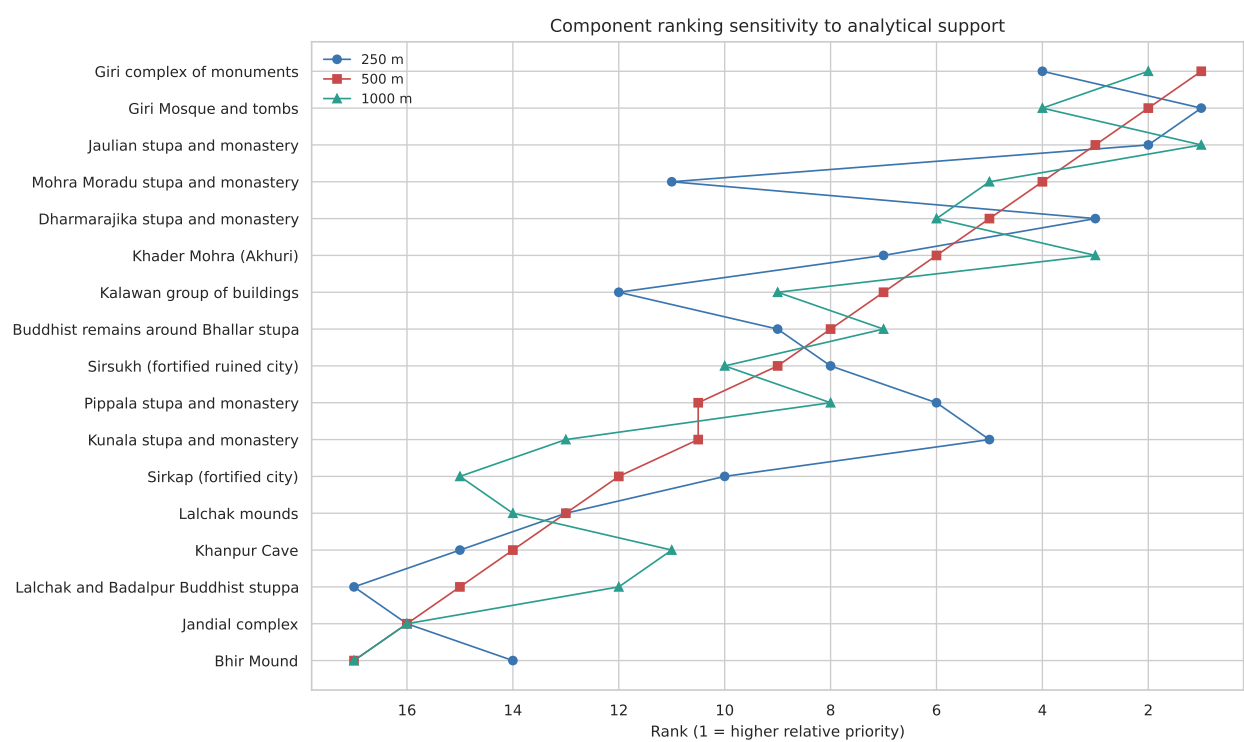


Figure S4: Component ranks at the three analytical neighbourhood radii.

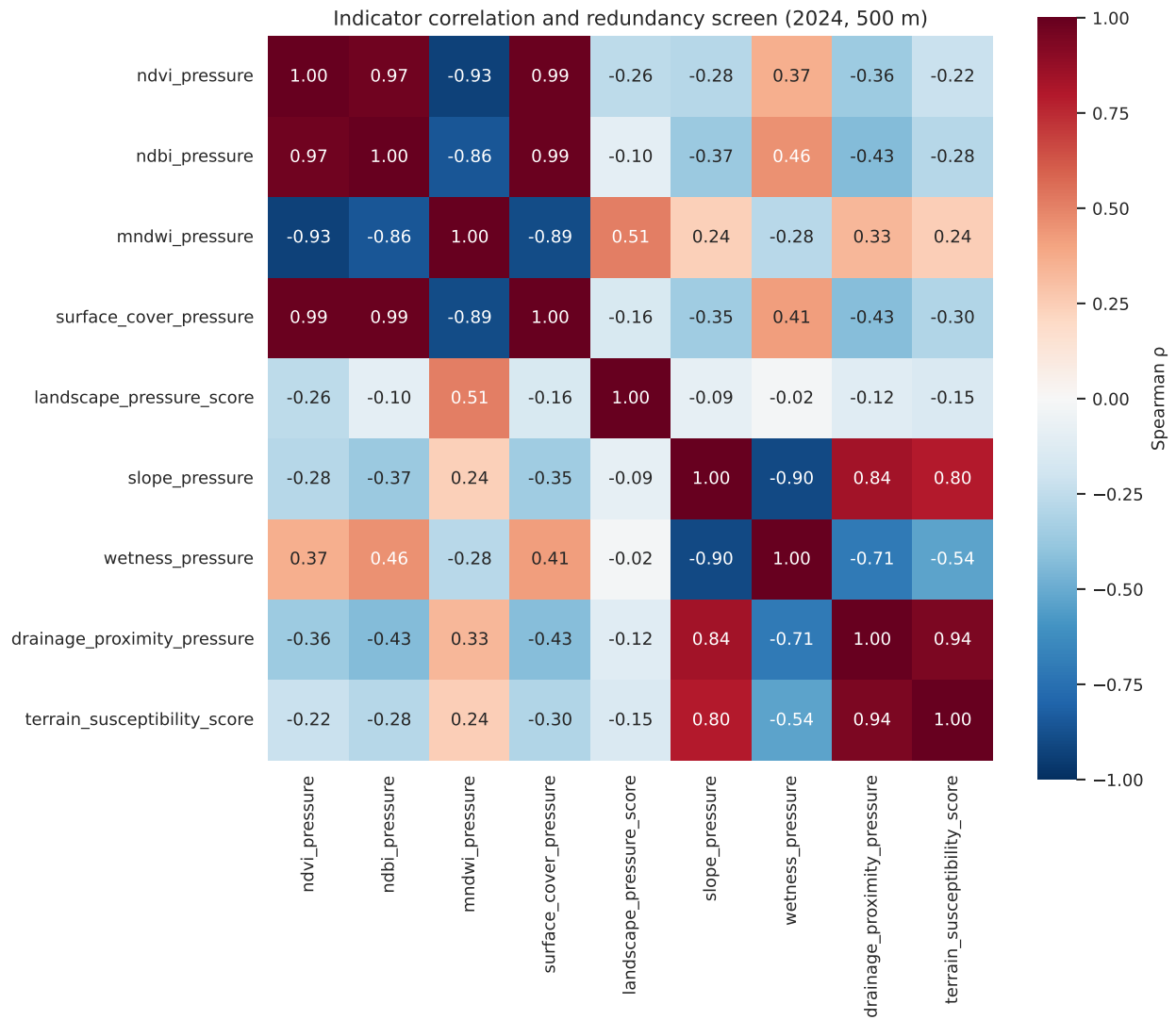


Figure S5: Dependence among component-level indicator ranks.

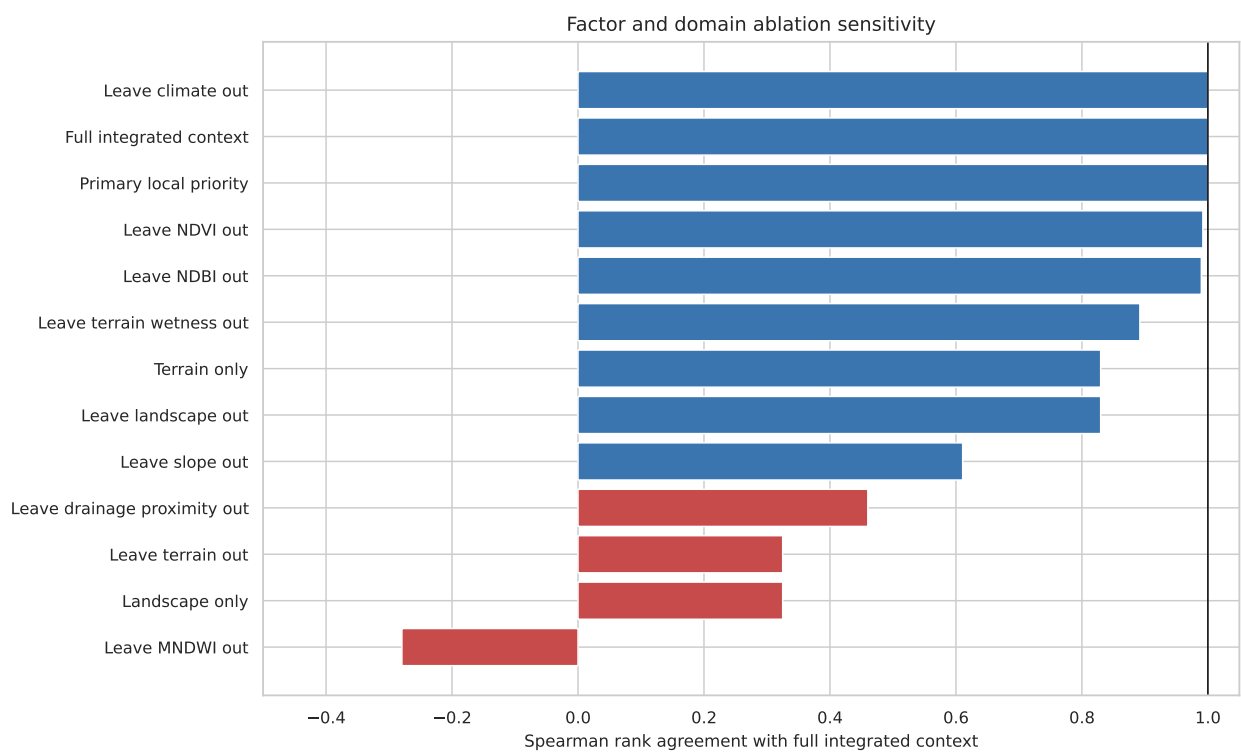


Figure S6: Archived factor/domain ablation. Changes in rank identify architectural sensitivity.